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Industrial Internship Report on: Java core- Banking Information System

Project Name: Digital Bank

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This report presents the work completed during my Winter Internship Program in Core Java. During this internship, I worked on developing a console-based project titled **"Digital Bank – Core Java Banking Information System."**The project includes basic banking features such as account creation, deposit, withdrawal, balance checking, and transaction history. Through this project, I applied core Java concepts like Object-Oriented Programming, ArrayList, loops, and conditional statements. This internship helped me improve my programming skills, logical thinking, and confidence in building structured Java applications independently. It provided practical exposure and strengthened my understanding of Core Java concepts. It gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface:

During this internship, I worked on developing a Core Java based project named "Digital Bank – Banking Information System." The project was developed step by step over four weeks, focusing on understanding Java fundamentals and implementing real-world banking operation

Internships play an important role in bridging the gap between theoretical knowledge and practical implementation. This internship helped me strengthen my Core Java concepts and gain hands-on experience in developing structured applications, which is essential for building a career in software development.

The main objective of the project was to design a simple banking system that performs operations such as account creation, deposit, withdrawal, balance inquiry, and transaction history tracking. The problem statement focused on implementing basic banking logic using Core Java concepts without using any database.

This internship opportunity provided by UpSkill Campus helped me gain practical exposure to Java programming. It allowed me to apply my academic knowledge in a real-world project environment and improve my technical and problem-solving skills.

The program was planned in a structured manner over four weeks.

- Week 1 – Understanding Core Java concepts and planning project structure
- Week 2 – Implementing account creation, deposit, and withdrawal features
- Week 3 – Adding transaction history and improving validation logic
- Week 4 – Testing, debugging, optimizing code, and preparing documentation

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Through this internship, I improved my understanding of Object-Oriented Programming, collections (ArrayList), exception handling, and logical implementation. I also learned how to debug errors, structure code properly, and document a technical project professionally.

I would like to thank UpSkill Campus for organizing this internship program and providing valuable learning resources. I am also thankful to the mentors and my college faculty for their guidance and support throughout the internship duration.

I would encourage my juniors and peers to participate in internships and practical projects along with their academic studies. Practical experience helps in understanding concepts more clearly and builds confidence in programming. Do not be afraid of making mistakes while coding, as debugging is an important part of learning. Stay consistent, keep practicing, and always be curious to learn new things.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA

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- It supports both cloud and on-premises deployments.

It has features to

• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine

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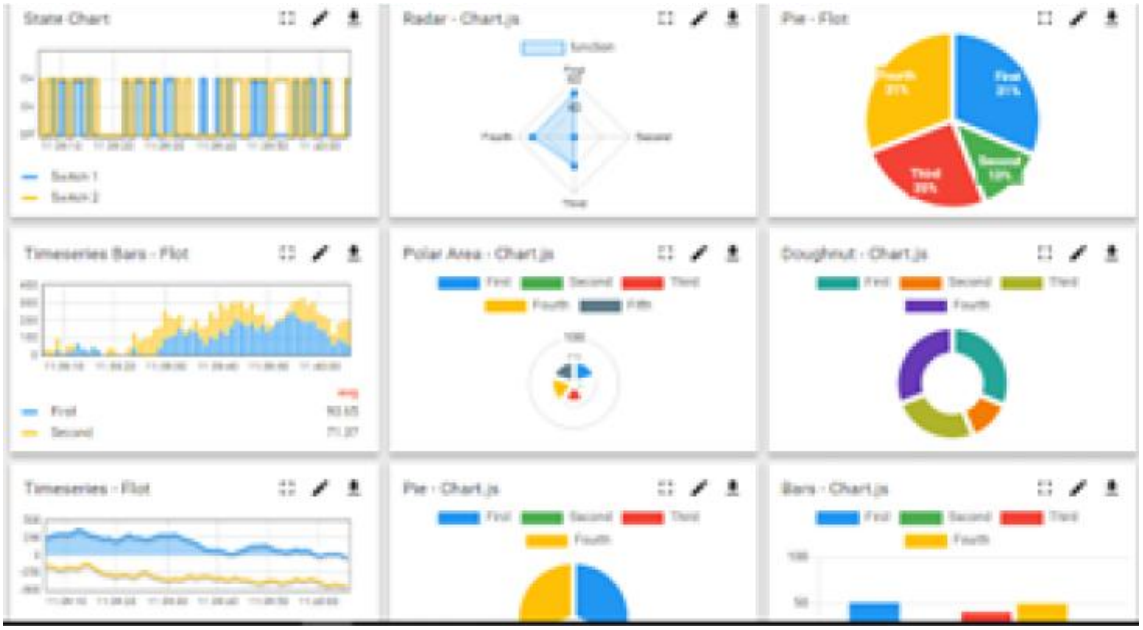

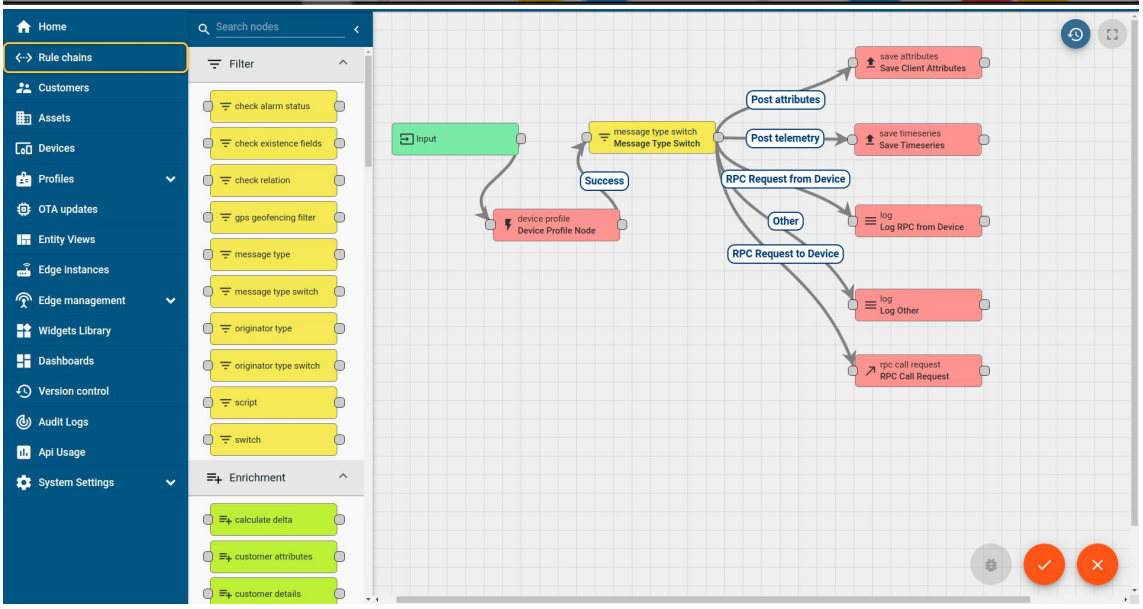

ii. Smart Factory Platform (FACTORY WATCH)

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Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

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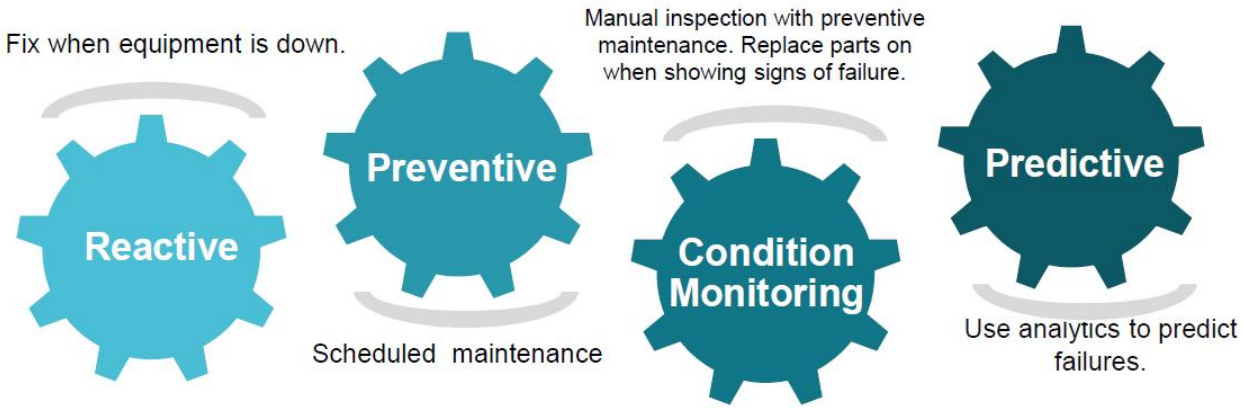
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

upSkill CAMPUS

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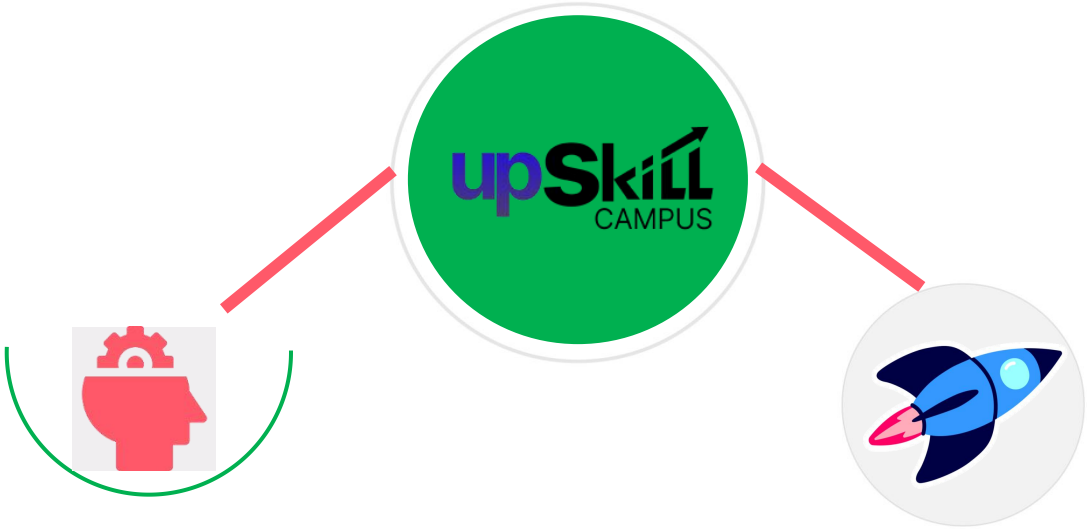
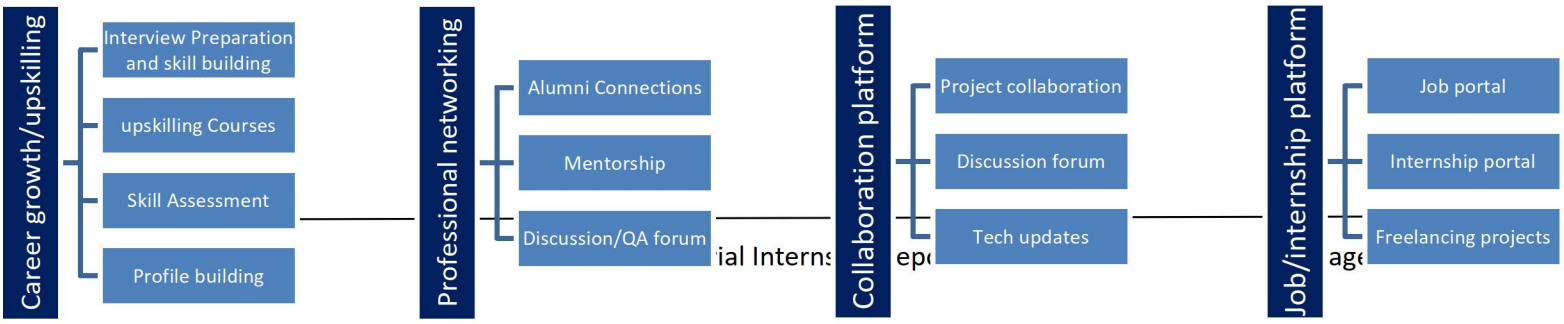


Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



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2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Oracle Java Documentation – https://docs.oracle.com/javase
[2] Core Java Programming Concepts – Online Tutorials and Study Materials
[3] UpSkill Campus Internship Learning Resources]

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2.6 Glossary

Terms	Acronym
Object oriented programming	oop
Java Development Kit	Java Development Kit
Integrated Development Environment	IDE
ArrayList	Java Collection Framework
Exception Handling	Error Handling Mechanism

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3 Problem Statement

In the assigned problem statement the objective of this project is to develop a console-based banking system using Core Java that performs basic banking operations such as account creation, deposit, withdrawal, balance inquiry, and transaction history management.

The problem focuses on implementing real-world banking logic using Object-Oriented Programming concepts without using a database. The system should handle multiple accounts, validate user input, and prevent invalid transactions such as insufficient balance withdrawal.

4 Existing and Proposed solution

Traditional banking systems use large-scale database-driven applications. Basic academic examples often demonstrate simple deposit and withdrawal logic but lack proper structure, validation, and transaction tracking.

Limitations:
• No structured class design
• No transaction history tracking
• Limited validation
• Not modular

The proposed solution is a structured Digital Bank system developed using Core Java principles.

Features:
• Account creation functionality
• Deposit and withdrawal with validation
• Balance inquiry
• Transaction history maintenance using ArrayList
• Exception handling for invalid inputs

The system follows modular class-based architecture to improve maintainability and readability.

4.1 Code submission (Github link) :

https://github.com/SnehalSukale27/upskillcampus/blob/main/BankingInformationSystem.java

4.2 Report submission (Github link) :

https://github.com/SnehalSukale27/upskillcampus/blob/main/BankingInformationSystem_Snehal_USC_UCT.pdf

5 Proposed Design/ Model

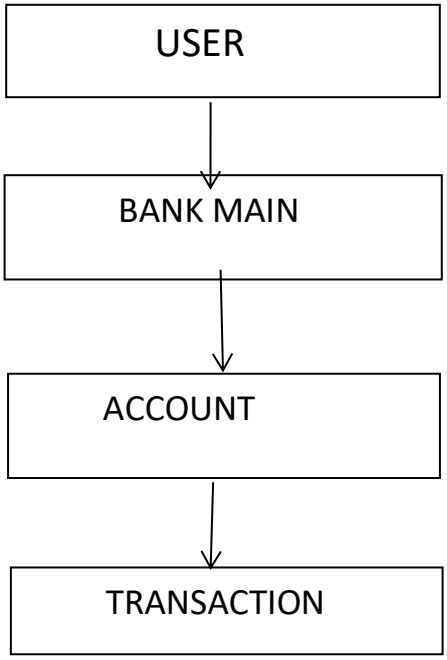

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.1 Interfaces

This project does not use Java Interfaces explicitly. However, it follows modular class-based architecture ensuring clear separation of responsibilities between account management and transaction handling.

6 Performance Test

The system was tested with multiple accounts and transaction scenarios to ensure proper functionality. Input validation and exception handling were tested to prevent invalid operations.

6.1 6.1 Test Plan / Test Cases

- Test account creation with valid inputs
- Test deposit with positive amount
- Test withdrawal with sufficient balance
- Test withdrawal with insufficient balance
- Test invalid menu option

6.2 6.2 Test Procedure

Manual testing was performed by executing different scenarios in the console. Multiple edge cases were tested to verify logic correctness.

6.3 6.3 Performance Outcome

The system successfully executed all banking operations without runtime errors. Validation mechanisms prevented invalid inputs and ensured reliable performance.

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7 My learnings

During this internship, I improved my understanding of Core Java concepts such as OOP, ArrayList, loops, and exception handling. I gained practical experience in designing structured applications and debugging logical errors. This project enhanced my problem-solving skills and confidence in Java programming.

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8 Future work scope

- Add file handling to store data permanently
- Implement database connectivity
- Add fund transfer functionality
- Develop GUI version using JavaFX or Swing
- Implement user authentication and security features